SmolVLM

Redefining Small and Efficient Multimodal Models

256M–2.2B
Parameters

< 1 GB RAM
SmolVLM-256M

300×
Smaller than Idefics-80B

Open Source
Weights · Data · Code · App

Hugging Face & Stanford University · arxiv.org/abs/2504.05299

A family of compact vision-language models achieving competitive performance
with dramatically lower memory footprints through strategic architectural optimizations.

Large VLMs Are Powerful but Impractical for Edge Deployment

THE CHALLENGE

- State-of-the-art VLMs require tens of gigabytes of GPU RAM
- Molmo-A1B-7B needs 27.7 GB — impractical for phones & edge devices
- Large models demand expensive cloud infrastructure for inference
- Mobile deployment blocked by parameter count and context overhead
- No fully open-source, sub-1GB multimodal model existed before SmolVLM

GPU RAM COMPARISON

Molmo-A1B-7B
27.7 GB
GPU RAM required

SmolVLM-256M
0.8 GB

SmolVLM-500M
1.2 GB

Key Insight: SmolVLM-256M uses only 0.8 GB GPU RAM — 35× less than Molmo-A1B-7B — enabling inference on smartphones and low-power edge devices.

SmolVLM Architecture

Image Splitting → Vision Encoder → Pixel Shuffle → Linear Projection → SmolLM2

1

Image Splitting

Split input into sub-images for detailed encoding

2

SigLIP Vision Encoder

93M param SigLIP-B/16 encodes visual features

3

Pixel Shuffle (r=4)

Space-to-depth compresses tokens 16× reduction

4

Linear Projection

Maps visual features into language model embedding space

5

SmolLM2 Backbone

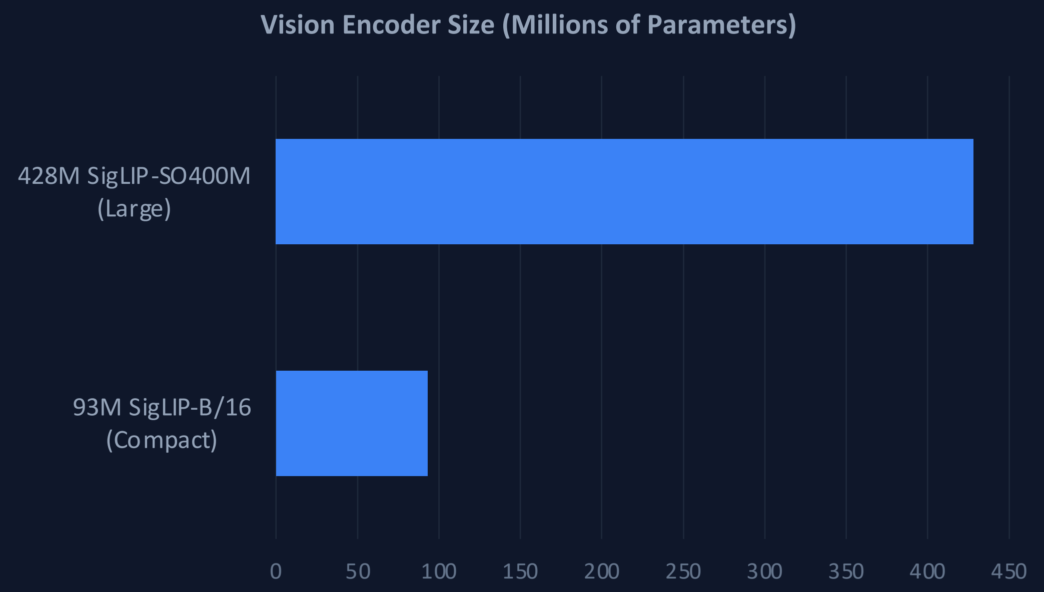
Language model generates text from fused tokens

ARCHITECTURE HIGHLIGHTS

- Video input: frames sampled and processed through the same pipeline as images
- Three model sizes share identical architecture — only backbone scale differs (135M / 360M / 1.7B LM)
- All weights, datasets, training code, and HuggingSnap mobile app are fully open-sourced

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1 — Balanced Encoder–LM Compute Allocation



✓ WINNER FOR SMALL LMs

93M SigLIP-B/16

Outperforms the 4.6× larger encoder when paired with the 135M SmolLM2 language model backbone.

CROSSOVER AT SCALE

428M SigLIP-SO400M

Only justifies its 10% parameter overhead at the 1.7B language model scale.

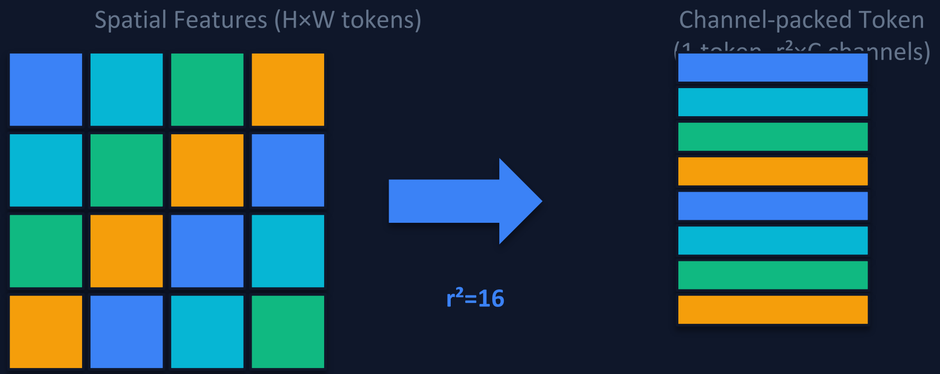
Design Principle

For compact VLMs, allocating more parameters to the language model backbone yields better multimodal reasoning than investing in a larger vision encoder. The optimal encoder–LM ratio shifts as total model size grows — a finding that challenges the default assumption of 'bigger encoder = better'.

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16×

Finding 3 — Token Compression via Space-to-Depth Rearrangement

PIXEL SHUFFLE MECHANISM



COMPRESSION FACTOR COMPARISON

r = 2 → 4× token reduction

Preferred for larger VLMs where attention overhead is manageable. Retains more spatial detail per token.

r = 4 → 16× token reduction ★

Optimal for compact VLMs — dramatically eases attention overhead and enables effective long-context modeling at small model scales. Used by SmolVLM-256M / 500M / 2.2B.

Impact

Pixel shuffle rearranges r×r spatial patches into channel dimensions, reducing sequence length by r^2. For small VLMs, the aggressive r=4 setting provides the best trade-off between information retention and computational efficiency, freeing up context budget for higher image resolution and multi-image inputs.

Extending Context from 2K to 16K Tokens Unlocks Higher Resolution

Finding 2 — Context Length Scaling via RoPE Base Adjustment

CONTEXT EXTENSION STRATEGY

Step 1

RoPE Base Adjustment

Increase from
10K → 273K

Step 2

Mixed-Length Fine-tuning

Train on both long
& short context data

Step 3

Model-Specific Context

2.2B → 16K tokens
256M/500M → 8K

WHY IT MATTERS

- Longer context enables processing more sub-images per input → higher effective image resolution
- Performance improves significantly when scaling from 2K to 16K tokens across all benchmarks
- Mixed long/short fine-tuning prevents catastrophic forgetting on short-context tasks
- Smaller models (256M, 500M) use 8K context as the optimal efficiency trade-off

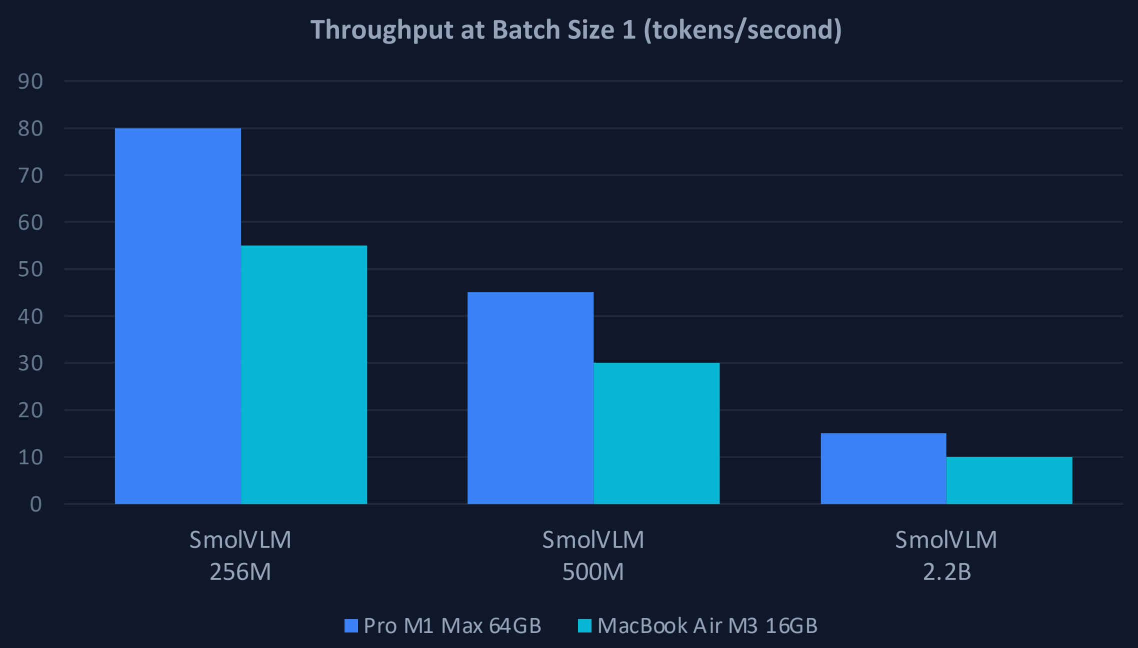
SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS Baseline
OCRBench	52.6%	61.0%	72.9%	54.7% Molmo-A1B-7B
AI2D	46.4%	59.2%	70.0%	71.0% Molmo-A1B-7B
ChartQA	55.6%	62.8%	68.7%	48.0% Molmo-A1B-7B
TextVQA	50.2%	60.2%	73.0%	61.5% Molmo-A1B-7B
DocVQA	58.3%	70.5%	80.0%	77.7% Molmo-A1B-7B
ScienceQA	73.8%	80.0%	89.6%	87.5% Molmo-A1B-7B
MMMU	29.0%	33.7%	42.0%	33.9% Molmo-A1B-7B
MathVista	35.9%	40.1%	51.5%	37.6% Molmo-A1B-7B
MMStar	34.6%	38.3%	46.0%	43.1% Molmo-A1B-7B
Video-MME	33.7%	42.2%	52.1%	45.0% InternVL2-2B
MLVU	40.6%	47.3%	55.2%	48.2% InternVL2-2B
AVERAGE	**44.0%**	**51.0%**	**59.8%**	—

GPU RAM → SmolVLM-256M: **0.8 GB** RAM | SmolVLM-500M: **1.2 GB** RAM | SmolVLM-2.2B: **4.9 GB** RAM | Molmo-A1B-7B: **27.7 GB** RAM

Real-Time Inference on Apple Silicon: Up to 80 Tokens/Second

On-Device Deployment via HuggingSnap iOS App — No Cloud Dependency



DEPLOYMENT HIGHLIGHTS

~80 tok/s
256M on M1 Max 64GB batch size 1

~1000 tok/s
256M on high-end devices batch size 64

~55 tok/s
256M on MacBook Air M3 16GB, batch size 1

HuggingSnap

Open-source iOS app processing photos and videos entirely on-device. No cloud dependency — all inference runs locally via optimized Apple Silicon kernels. SmolVLM-256M fits comfortably within iPhone memory constraints.

Key Takeaways: Design Principles for Efficient Multimodal Models

1 Balance Encoder–LM Compute

Smaller vision encoders (93M SigLIP) outperform larger ones (428M) for compact LMs. Invest parameters in the language backbone, not the vision encoder.

2 Extend Context Length

Scaling from 2K → 16K tokens via RoPE base adjustment (10K → 273K) unlocks higher effective image resolution and significantly improves benchmark scores.

3 Aggressive Token Compression

Pixel shuffle at r=4 provides 16× token reduction — the optimal trade-off for small VLMs between information retention and computational efficiency.

4 Sub-1GB Inference Is Achievable

SmolVLM-256M runs at 0.8 GB GPU RAM and outperforms the 300× larger Idefics-80B. Architectural efficiency matters more than raw parameter scale.

5 Open Source Everything

All weights, datasets, training code, and the HuggingSnap mobile app are released — enabling reproducibility and accelerating on-device AI research.

Implication for On-Device AI: SmolVLM demonstrates that careful co-design of vision encoding, token compression, and context scaling can deliver multimodal AI that fits in a smartphone's memory budget — with fully competitive accuracy.